# 60A, 100V Trench Schottky Surface Mount Rectifier

## FEATURES

- AEC-Q101 qualified
- Low power loss, high efficiency
- Ideal for automated placement
- High surge current capability
- RoHS compliant
- Halogen-free

## APPLICATIONS

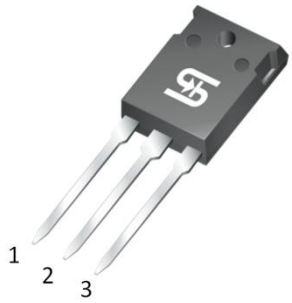
- Low voltage, high frequency
- DC/DC converter
- Freewheeling diodes
- Reverse battery protection
- Car lighting

## MECHANICAL DATA

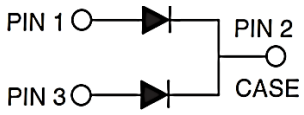
- Case: TO-247AD
- Molding compound meets UL 94V-0 flammability rating
- Terminal: Matte tin plated leads, solderable per J-STD-002
- Meet JESD 201 class 2 whisker test
- Polarity: As marked
- Weight: 6.44g (approximately)

| KEY PARAMETERS | | |
|---|---|---|
| **PARAMETER** | **VALUE** | **UNIT** |
| $I_F$ | 60 | A |
| $V_{RRM}$ | 100 | V |
| $I_{FSM}$ | 345 | A |
| $T_{J\ MAX}$ | 175 | °C |
| Package | TO-247AD | |
| Configuration | Common cathode | |

Pb-Free RoHS COMPLIANT HALOGEN FREE




**TO-247AD**




| ABSOLUTE MAXIMUM RATINGS ($T_A$ = 25°C unless otherwise noted) | | | |
|---|---|---|---|
| **PARAMETER** | **SYMBOL** | **VALUE** | **UNIT** |
| Repetitive peak reverse voltage | $V_{RRM}$ | 100 | V |
| Reverse voltage, total rms value | $V_{R(RMS)}$ | 70 | V |
| Forward current, per leg / per device | $I_F$ | 30 / 60 | A |
| Peak forward surge current, 8.3ms single half sine-wave superimposed on rated load | $I_{FSM}$ | 345 | A |
| Junction temperature | $T_J$ | - 55 to +175 | °C |
| Storage temperature | $T_{STG}$ | - 55 to +175 | °C |

## THERMAL PERFORMANCE

| PARAMETER | SYMBOL | TYP | UNIT |
|---|---|---|---|
| Junction-to-lead thermal resistance per leg | $R_{\Theta JL}$ | 1.2 | °C/W |
| Junction-to-ambient thermal resistance per leg | $R_{\Theta JA}$ | 5.1 | °C/W |
| Junction-to-case thermal resistance per leg | $R_{\Theta JC}$ | 1.3 | °C/W |

**Thermal Performance Note:** Mounted on Heat sink with 4" x 6" x 0.25" Al-Plate

## ELECTRICAL SPECIFICATIONS ($T_A$ = 25°C unless otherwise noted)

| PARAMETER | CONDITIONS | SYMBOL | TYP | MAX | UNIT |
|---|---|---|---|---|---|
| Forward voltage per leg(1) | $I_F$ = 15A, $T_J$ = 25°C | $V_F$ | 0.68 | - | V |
| | $I_F$ = 30A, $T_J$ = 25°C | | 0.78 | 0.84 | V |
| | $I_F$ = 15A, $T_J$ = 125°C | | 0.57 | - | V |
| | $I_F$ = 30A, $T_J$ = 125°C | | 0.67 | 0.73 | V |
| Reverse current @ rated $V_R$ per leg(2) | $T_J$ = 25°C | $I_R$ | - | 30 | μA |
| | $T_J$ = 125°C | | - | 10 | mA |
| Junction capacitance per leg | 1MHz, $V_R$ = 4.0V | $C_J$ | 1160 | - | pF |

**Notes:**

1. Pulse test with PW = 0.3ms
2. Pulse test with PW = 30ms

## ORDERING INFORMATION

| ORDERING CODE | PACKAGE | PACKING |
|---|---|---|
| TSH60H100CH | TO-247AD | 30 / Tube |

## CHARACTERISTICS CURVES

($T_A$ = 25°C unless otherwise noted)

**Fig.1 Forward Current Derating Curve**

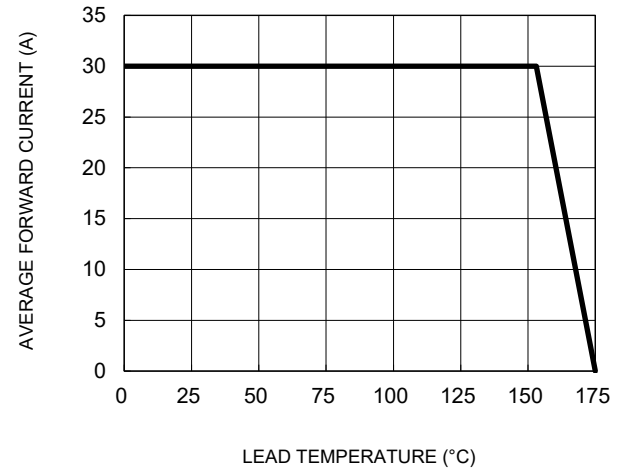



**Fig.2 Typical Junction Capacitance**

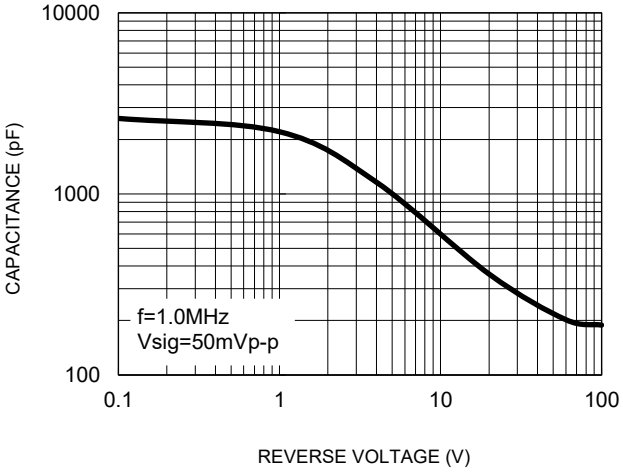



**Fig.3 Typical Reverse Characteristics**

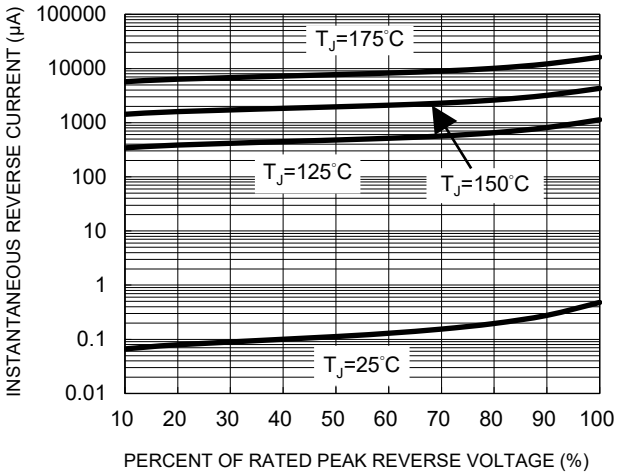



**Fig.4 Typical Forward Characteristics**

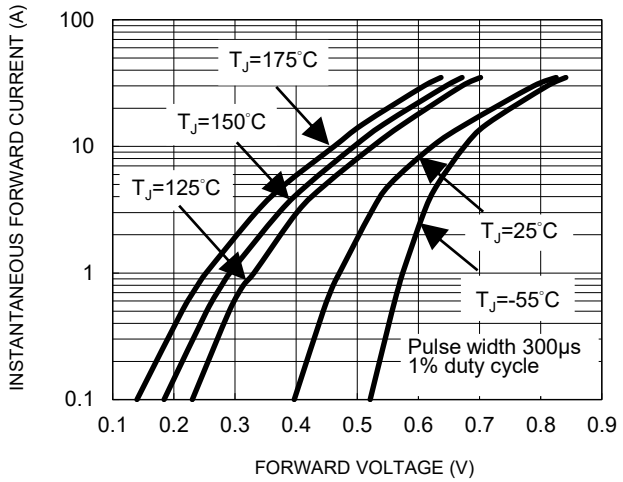
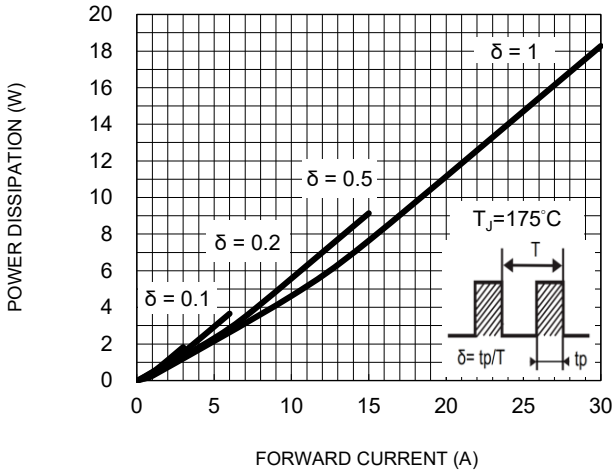


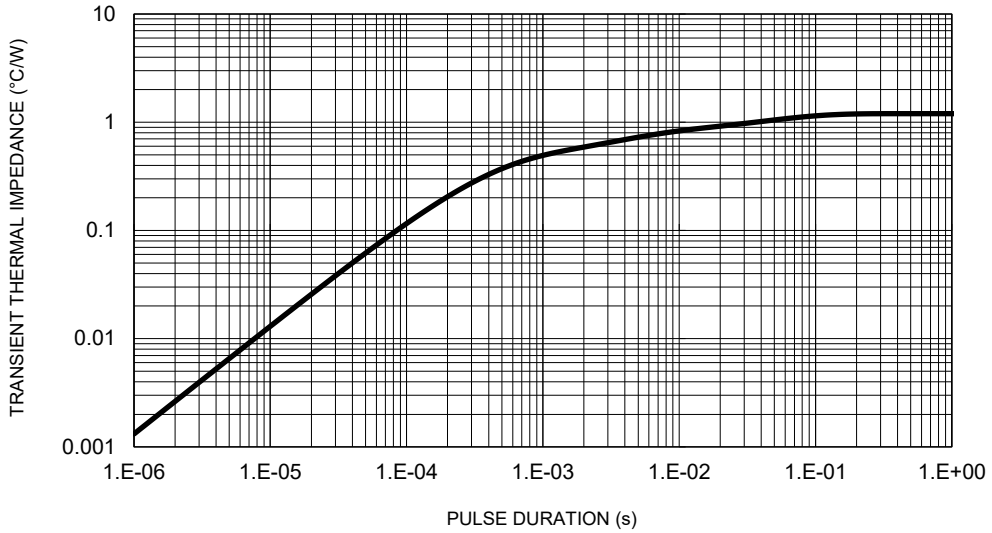

**Fig.5 Typical Forward Power Dissipation vs. Forward Current**



## CHARACTERISTICS CURVES

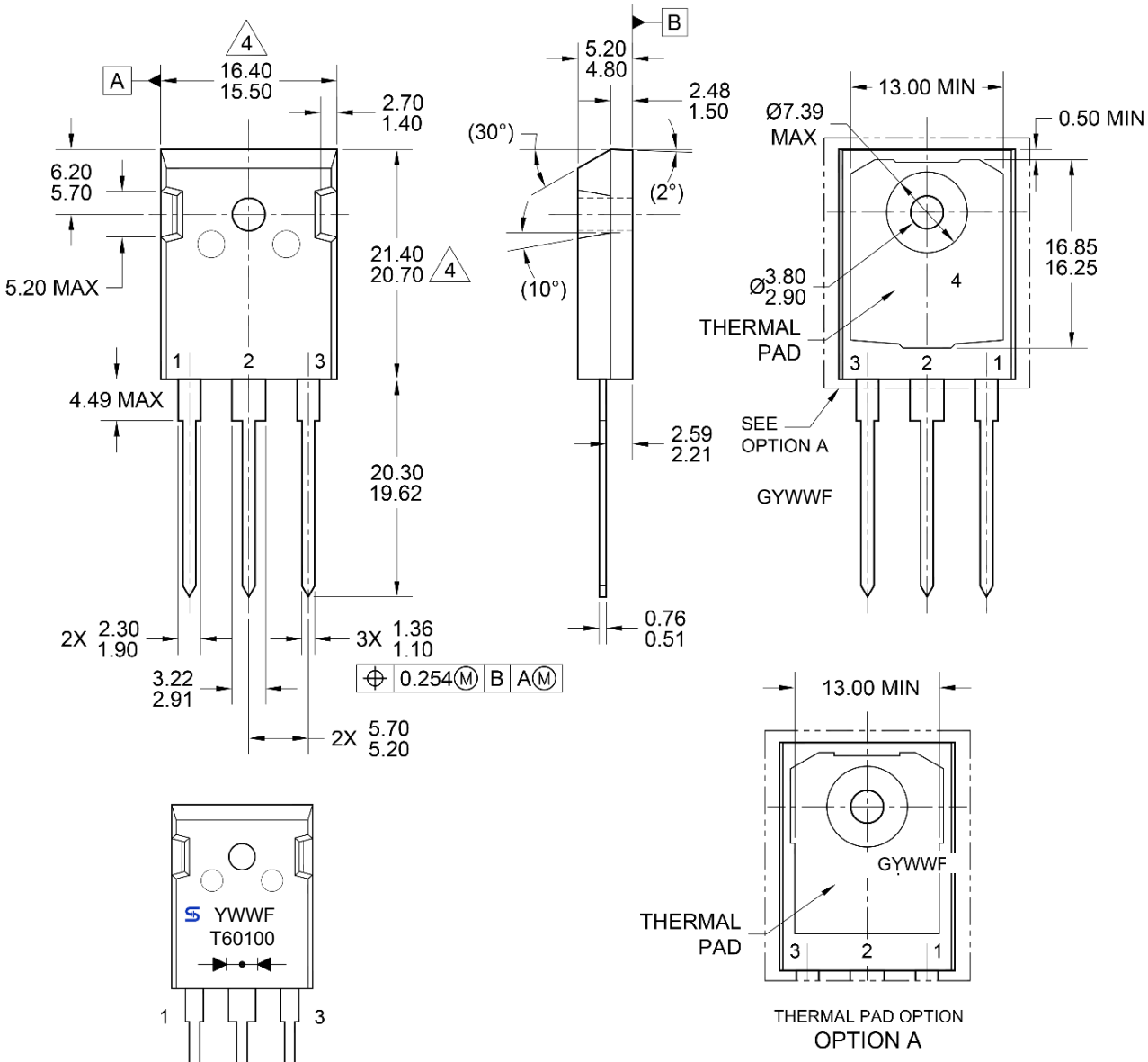
($T_A$ = 25°C unless otherwise noted)

**Fig.6 Typical Transient Thermal Characteristics**



## PACKAGE OUTLINE DIMENSIONS

**TO-247AD**




MARKING DIAGRAM

YWW = Date code
F = Factory code

NOTES: UNLESS OTHERWISE SPECIFIED

1. ALL DIMENSIONS ARE IN MILLIMETERS.
2. DIMENSIONING AND TOLERANCING PER ASME Y14.5M-1994.
3. PACKAGE OUTLINE REFERENCE: JEDEC TO-247, VARIATION AD, ISSUE E.
4. DIMENSIONS D AND E DO NOT INCLUDE MOLD FLASH. THESE DIMENSIONS ARE MEASURED AT THE OUTERMOST EXTREME OF THE PLASTIC BODY.
5. DWG NO. REF: HQ2SD07-TO247AD-071 REV B.

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